Throughout this worksheet, assume $f(t)$ and $g(t)$ are continuous.

Integral at One Point Only

$$\int_a^a f(t)\,dt = 0.$$

Constant Multiple Rule

Let c be a constant, then $\int_a^b cf(t)\,dt = c\int_a^b f(t)\,dt$.

Reversing the Direction of an Integral

$$\int_b^a f(t)\,dt = -\int_a^b f(t)\,dt$$

Sum Rule

$$\int_a^b f(t) + g(t)\ dt = \int_a^b f(t)\ dt + \int_a^b g(t)\ dt.$$

Difference Rule

$$\int_a^b f(t) - g(t)\ dt = \int_a^b f(t)\ dt - \int_a^b g(t)\ dt.$$

Splitting Domains of Integration

$$\int_a^b f(t)\,dt = \int_a^c f(t)\,dt + \int_c^b f(t)\,dt.$$

Bounding Integrals

If $m \leq f(t) \leq M$ for all values of t in $[a, b]$ then $m(b-a) \leq \int_a^b f(t)\, dt \leq M(b-a)$.

Integrals of Smaller and Larger Functions

If $f(t) \leq g(t)$ for all values of t in $[a, b]$ then $\int_a^b f(t)\, dt \leq \int_a^b g(t)\, dt$

Another Riemann Sums Exercise

Consider the Riemann sum $\sum_{k=0}^{499} \left(5 - (2 + 0.006k)^2\right) \Delta x$.

1. What are the values of Δx and a? (Hint: somewhere in that expression is $a + k\Delta x$. □Regrade Find it to get values for a and Δx.)

2. What definite integral is being approximated? (Hint: you will need to find b, and □Regrade $f(x)$.)

3. Is this Riemann sum an overestimate of the integral, an underestimate of it, or is it □Regrade impossible to tell? Explain your answer.

4. Suppose we write another Riemann sum to approximate the same definite integral. □Regrade This new Riemann sum has the form

$$\sum_{k=0}^{9999} (5 - (2 + _____) \Delta x.$$

   (a) What is the new value of Δx?

   (b) Fill in the blank in the new Riemann sum.

   (c) Is this new Riemann sum an overestimate or an underestimate of the definite integral? Explain.

   (d) Is this new Riemann sum smaller or larger than the first one? Explain.